Exercises in *Fourier Analysis on Number Fields*

Solutions to the exercises of Ramakrishnan–Valenza

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Contents

1 Tate's Thesis and Applications **3**

1 Topological Groups

2 Some Representation Theory

3 Duality for Locally Compact Abelian Groups

4 The Structure of Arithmetic Fields

5 Adeles, Ideles, and the Class Groups

6 A Quick Tour of Class Field Theory

7 Tate's Thesis and Applications

Exercise 7.1. Let K be a local field and let ψ be a nontrivial unitary character on K; that is, a continuous, nontrivial homomorphism from K as an additive group into S^1, the circle group. For each $a \in K$, define $\psi_a \colon K \to S^1$ by

$$\psi_a(x) = \psi(ax).$$

(a) Show that ψ_a is again a unitary character of K, and that ψ_a is trivial if and only if $a = 0$.

(b) Show that $a \mapsto \psi_a$ defines a continuous, injective homomorphism α_ψ of K into its Pontryagin dual $\widehat{K}$.

(c) Show that the image of α_ψ is dense in $\widehat{K}$. [*Hint*: Show that $\psi_a(b) = 1$ for all $a \in K$ if and only if $b = 0$.]

(d) Show that α_ψ is bicontinuous.

(e) Show that $\alpha_\psi(K)$ is a complete, hence closed, subgroup of $\widehat{K}$. Conclude that α_ψ is an isomorphism of topological groups.

Exercise 7.2. Let K be a non-Archimedean local field with ring of integers $\mathfrak{o}_K$ and maximal ideal P. Fix a nontrivial unitary character ψ of K. Let P^m be the conductor of ψ; that is, the largest fractional ideal P^m on which ψ is trivial. Let α_ψ, as defined in Exercise 1.1, be the isomorphism between K and its Pontryagin dual.

(a) Show that via α_ψ, P^m identifies with the Pontryagin dual of $\widehat{\mathfrak{o}}_K$. (This is also the inverse different of K, which soon makes its appearance.)

(b) Let $n \geq 1$ and $x \in K$. Show that $xP^n \subseteq \ker(\psi)$ if and only if $x \in P^{m-n}$.

Bibliography

[RV99] D. Ramakrishnan and R. J. Valenza, *Fourier Analysis on Number Fields*, Graduate Texts in Mathematics **186**, Springer-Verlag, New York, 1999.